Sensitivity-Engine Derivative Surrogates: Greek-Regularized and Boundary-Aware Supervision*

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Abstract

This paper presents a derivative-surrogate learning framework built on an earlier sensitivity-engine sequence consisting of BSE, DSE, NDSE, and DBSE, and studies that framework in a first completed empirical implementation. The purpose is not to displace established pricing theory or classical numerical methods, but to transfer pricing structure into supervision. Earlier papers in the sequence provide exact or semi-structured labels for prices, first-order Greeks, mixed second-order Greeks, digital weights, conditional-law blocks, and boundary-sensitive distributional terms under deterministic or stochastic discounting. The present paper organizes those objects into a unified learning framework for derivative pricing, hedging, and local risk approximation.

Methodologically, the paper formulates structured losses and architectures that train on prices together with selected sensitivities and, where appropriate, with boundary-aware auxiliary targets. Greek-regularized losses are interpreted as weighted Sobolev-type objectives. Multi-head decompositions are matched to the product structure already present in DSE and NDSE. Near nonsmooth exercise regions, DBSE motivates mollified targets, adaptive boundary sampling, and auxiliary digital heads. Under stochastic discounting, NDSE motivates measure-aware labels and low-dimensional term-structure encodings that preserve the distinction between the spot leg and strike leg. The framework is therefore broader than the present empirical implementation, while the free-data study reported here tests its price-Greek and boundary-aware components in a controlled setting.

The empirical study uses public S&P 500, Treasury, and implied-volatility data to construct daily market states, then generates synthetic option labels from a Black–Scholes–Merton teacher on a fixed maturity-moneyness grid. Three surrogate specifications are compared: a price-only model, a Greek-regularized model, and a boundary-aware Greek-regularized model. Relative to the price-only baseline, structured supervision improves Delta and Gamma accuracy and reduces local hedging dispersion and tail loss. In the completed empirical study, the boundary-aware specification (Model C) delivers the most favorable overall performance among the three specifications within the present design, with the best global price RMSE, local price RMSE, Delta accuracy, and hedging-dispersion and tail-risk metrics. The Greek-regularized non-boundary specification (Model B) remains a useful conservative benchmark because it retains the best Gamma accuracy.

*This paper continues *Dirac Blocks for Sensitivity Engines* (DBSE) [18], which continues *Numeraire-Based Distributional Sensitivity Engine* (NDSE) [17], *Distributional Sensitivity Engine* (DSE) [16], and *BSM Sensitivity Engine* (BSE) [15]. The empirical implementation accompanying the present paper is available at `https://github.com/DavidHShen/Sensitivity-Engine_Derivative_Surrogates`.

The evidence is therefore consistent with the view that sensitivity-engine-guided learning is most useful through local risk fidelity and hedge robustness, and that boundary-aware supervision can be effective when tuned carefully rather than imposed too aggressively.

Keywords: derivative surrogates, financial machine learning, Greek regularization, boundary-aware learning, local hedging, stochastic discounting, change of numeraire

JEL: G12, G13, C45, C63

1 Introduction

Machine-learning surrogates are increasingly used in derivative pricing because a trained model can deliver low-latency evaluation across large state spaces. Recent work on deep hedging and neural surrogate calibration indicates both the practical demand for fast evaluation and the importance of aligning learning systems with pricing and hedging structure rather than relying only on generic interpolation [8, 9]. In many practical settings, however, price accuracy alone is not enough. Hedging, scenario analysis, stress testing, calibration, and risk aggregation depend on stable first- and second-order sensitivities. A surrogate that interpolates option values well but produces unstable Delta or Gamma can still be unsuitable for risk-sensitive use. The difficulty is especially pronounced near short-dated near-the-money regions and other nonsmooth payoff layers, where naive price-only learning can underresolve the local geometry that actually drives hedge performance.

The earlier sensitivity-engine sequence suggests a specific response to that problem. *BSM Sensitivity Engine* (BSE) derives implementation-friendly first- and mixed second-order Black–Scholes–Merton (BSM) sensitivities from a compact generator representation [15]. *Distributional Sensitivity Engine* (DSE) extends that viewpoint beyond the closed-form setting by rewriting prices and Greeks through two-measure digital weights and conditional-law blocks under deterministic discounting [16]. *Numeraire-Based Distributional Sensitivity Engine* (NDSE) carries the same logic into stochastic discounting through numeraire-matched measures, thereby preserving the basic engine structure while accommodating bond normalization and forward-measure terms [3, 17]. *Dirac Blocks for Sensitivity Engines* (DBSE) isolates the boundary layer generated by digital blocks, records the resulting Dirac and δ' structure, and links those objects to mollifier and likelihood-ratio Monte Carlo estimators [18]. Taken together, these earlier contributions provide a hierarchy of supervisory objects for derivative-focused learning that extends beyond pricing formulas alone.

The present paper sets out that learning layer and examines it in a controlled first empirical implementation. The central claim is intentionally narrow. Structured pricing and sensitivity engines can serve as numerically stable teachers and regularizers for derivative machine-learning systems, especially when the task requires accurate prices, first-order Greeks, mixed second-order Greeks, and stable behavior near nonsmooth boundaries or under stochastic discounting. This should not be read as a claim that machine learning ought to replace pricing theory. The narrower claim is that pricing structure can be exploited directly inside the learning objective.

The empirical findings help determine how that claim should be stated. The completed free-data study compares a price-only surrogate (Model A), a Greek-regularized surrogate (Model B), and a boundary-aware Greek-regularized surrogate (Model C). Relative to Model A, both structured models improve derivative accuracy and local hedging stability. Model C is the strongest overall specification in the present design, while Model B remains the strongest Gamma specification. The main value of sensitivity-engine-guided learning appears in local risk fidelity and hedge robustness, and boundary-aware supervision appears useful when tuned carefully rather than imposed mechanically.

The paper is organized around three main contributions. First, it reformulates the results of the earlier sensitivity-engine sequence as a hierarchy of supervisory objects for learning, including exact labels, probabilistic weights, decomposition primitives, and boundary-sensitive auxiliary targets. Second, it formulates a unified learning framework in which price, Greek, decomposition, and boundary information can be combined within a single objective. Third, it reports a completed free-data empirical study based on daily S&P 500 market states and a synthetic BSM teacher. That empirical study shows that structured supervision can improve derivative-surrogate performance not only through Greek accuracy but also through local price fidelity and hedging robustness, while preserving a metric-dependent interpretation of the model ranking.

2 Engine background from the earlier sensitivity-engine sequence

This section summarizes only the ingredients needed for the present learning framework. The earlier components of the sequence are not re-derived in full.

2.1 Exact baseline labels from BSE

BSE [15] provides a compact derivation of first-order and mixed second-order BSM sensitivities by differentiating the classical option-pricing formulas of BSM with respect to an arbitrary scalar parameter and then organizing the result into reusable weight channels [1, 2]. Two features matter for the present paper.

First, the paper yields essentially noise-free labels for prices and Greeks over the BSM parameter space. Those labels are valuable for learning because they make it possible to study approximation error without contamination from Monte Carlo noise or PDE discretization. Second, BSE already has an implementation-friendly structure based on common primitives and a universal correction channel. That makes it a natural teacher in baseline supervised experiments and a clean reference point for diagnosing whether a learning architecture reproduces both level behavior and local geometry.

2.2 Probabilistic block structure from DSE

DSE [16] extends the engine viewpoint beyond the closed-form setting. For deterministic discounting, the in-the-money event is rewritten through a Doleans–Dade exponential and a Girsanov tilt, producing a two-weight price form

$$V_t = w_1 a_1 + w_2 a_2, \tag{1}$$

where V_t denotes the time-t derivative value, the prefactors a_1, a_2 are $\mathcal{F}_t$-measurable primitive quantities, and the weights w_1, w_2 are conditional digital objects under matched measures. Differentiation then yields a four-input first-derivative engine, and mixed partials retain a structured "primitive plus correction" form analogous to the BSE setting.

For machine learning, the significance of DSE is that supervision no longer needs to be limited to scalar prices. The teacher can supply digital probabilities, density-like derivative blocks, and mixed-sensitivity correction channels. These objects are interpretable and numerically meaningful, and they can be attached either directly to a loss function or indirectly through architectural design.

2.3 Stochastic-discount extension from NDSE

NDSE [3, 17] extends the same logic to stochastic discounting. The discount factor is normalized by the bond price $P(t,T) = \mathbb{E}_t^{\mathbb{Q}}[D_{t,T}]$, producing a modified boundary variable $\widetilde{E}_{t,T}$ and an $\mathcal{F}_t$-measurable exercise threshold. The spot leg is retained under a Doleans tilt measure $\mathbb{Q}^+$, while the strike leg is expressed under the T-forward measure $\mathbb{Q}^T$, or more generally under a suitable numeraire measure. The basic engine structure survives this extension. The price keeps a two-weight form and first derivatives retain a four-input contraction representation.

For learning, this extension matters because a deterministic-rate surrogate setting does not fully capture the measure and discounting structure required when rate dynamics are stochastic. NDSE provides labels that already respect the pricing measure appropriate to each leg. As a consequence, a learning system can be trained on stochastic-discount data without collapsing the problem back into a deterministic-rate approximation.

2.4 Boundary layer from DBSE

DBSE [18] focuses on digital blocks of the generic form

$$B(m;\theta) = \mathbb{E}_t^{\mathsf{M}_\theta}\left[\mathbf{1}_{\{X(\theta)<m(\theta)\}}\right], \tag{2}$$

where $B(m;\theta)$ denotes the conditional digital block, θ is the parameter vector, $X(\theta)$ is the boundary-relevant random variable, $m(\theta)$ is the exercise threshold, and M_θ is the parameter-dependent pricing or auxiliary measure. Thus both the boundary pair (X, m) and possibly the target measure may depend on parameters. Differentiation generates weighted Dirac terms at first order and δ'-type terms at second order. When the target measure itself depends on parameters, likelihood-ratio corrections enter on top of the boundary terms. The paper also records the corresponding Monte Carlo implementation layer through mollifiers and bandwidth guidance, building on the standard simulation-based view of Greek estimation [4, 5].

The machine-learning consequence is straightforward. Price-only objectives do not directly encode this nonsmooth boundary geometry. Therefore, a surrogate trained only on scalar prices may achieve acceptable global level fit while failing precisely where digital terms dominate local sensitivities. DBSE provides a principled route for targeted supervision near those regions.

2.5 Unified interpretation

Taken together, the four papers suggest a layered supervisory hierarchy:

Layer	Machine-learning role
BSE	Exact price and Greek labels in the BSM baseline; error diagnosis without numerical label noise
DSE	Deterministic-discount digital blocks and sensitivity labels beyond closed form; interpretable intermediate targets
NDSE	Stochastic-discount labels under numeraire-matched measures; supervision with explicit rate-curve dependence
DBSE	Boundary-aware targets, mollified labels, and likelihood-ratio corrections near nonsmooth regions

The rest of the paper sets out a learning framework that uses this hierarchy deliberately rather than treating the earlier papers merely as offline label generators.

3 Learning problems and supervisory objects

3.1 Input space and target families

Let $\Theta \subset \mathbb{R}^d$ denote the joint space of market, model, and contract descriptors. A generic input vector may be written as

$$\theta = (\theta^{\text{mkt}}, \theta^{\text{mdl}}, \theta^{\text{ctrt}}), \tag{3}$$

where the superscripts separate market-state inputs, model parameters, and contract descriptors. Representative coordinates include spot, strike, maturity, dividend descriptors, volatility parameters, factor loadings, and rate-curve descriptors. For stochastic-discount settings, a low-dimensional curve representation may be appended:

$$\theta^{\text{curve}} = (c_1, \dots, c_q), \tag{4}$$

where (c_j) may denote principal-component coordinates of a zero-coupon curve or a reduced factor vector from an HJM-, LMM-, or generic-term-structure model.

The teacher engine delivers one or more of the following target families:

$$V(\theta) \quad \text{price}, \tag{5}$$

$$G(\theta) = \big(\partial_{\theta_i} V(\theta)\big)_{i \in \mathcal{I}} \quad \text{selected first-order Greeks}, \tag{6}$$

$$H(\theta) = \big(\partial^2_{\theta_i \theta_j} V(\theta)\big)_{(i,j) \in \mathcal{J}} \quad \text{selected second-order Greeks}, \tag{7}$$

$$W(\theta) \quad \text{digital / conditional-law / boundary auxiliary targets}. \tag{8}$$

In this display, V is the scalar price target, G collects selected first-order derivative targets, H collects selected second-order derivative targets, and W collects non-price auxiliary targets such as digital probabilities or boundary terms. The index sets $\mathcal{I}$ and $\mathcal{J}$ are task dependent; $\mathcal{I}$ selects first-order coordinates such as spot or volatility sensitivities, while $\mathcal{J}$ selects the second-order coordinates actually needed for the task. In practice, it is seldom necessary to supervise the full Hessian.

3.2 Primary application tasks

Four task classes are emphasized.

T1. Surrogate pricing. Approximate $\theta \mapsto V(\theta)$ together with selected sensitivities.

T2. Calibration support. Learn maps from option surfaces or compressed surface features to latent model parameters, with sensitivity information serving as a stabilizer or auxiliary target.

T3. Hedging and scenario-risk approximation. Approximate prices and sensitivities jointly, with evaluation based partly on hedging P&L or local shock error.

T4. Stochastic-rate generalization. Learn in a joint state space that includes curve information and assess robustness under shifted rate regimes.

The first task is the cleanest entry point because it is easiest to diagnose and least confounded by inverse-problem instability. The remaining tasks become natural once the surrogate can already reproduce prices and sensitivities reliably.

3.3 Learning architectures

The framework is compatible with several model classes. In the present implementation, the empirical study uses multilayer perceptrons because they are easy to differentiate and keep the comparison with the analytic teacher transparent. More elaborate architectures, including transformer-style encoders or operator-learning designs for surface-to-parameter tasks, remain compatible with the framework but are not needed for the present paper.

Three architectural patterns are particularly relevant.

a) **Single-output differentiable surrogates,** where a single network $\widehat{V}_\phi(\theta)$ is differentiated by automatic differentiation to obtain surrogate Greeks.

b) **Direct multi-head surrogates,** where the network outputs $(\widehat{V}_\phi, \widehat{G}_\phi, \widehat{H}_\phi, \widehat{W}_\phi)$ explicitly.

c) **Decomposition-aware surrogates,** where the architecture mirrors a product form such as $a_1 w_1 + a_2 w_2$ and attaches supervision to individual heads.

The third design is distinctive because it incorporates the structure already exposed by DSE and NDSE.

4 Greek-regularized learning

4.1 Price-only benchmark and structured losses

Let μ denote a sampling distribution on Θ, let ϕ denote trainable surrogate parameters, and let $V(\theta)$ and $\widehat{V}_\phi(\theta)$ denote the teacher price and model-implied surrogate price at input θ. A price-only benchmark is trained by minimizing

$$\mathcal{R}_{\text{price}}(\phi) = \int_\Theta \ell_V\big(\widehat{V}_\phi(\theta), V(\theta)\big)\, \mu(\,\mathrm{d}\theta). \tag{9}$$

Here ℓ_V is the scalar price-loss function and $\mu(\,\mathrm{d}\theta)$ indicates averaging over the empirical or design distribution of inputs. For squared loss,

$$\ell_V(a, b) = |a - b|^2, \tag{10}$$

where a and b denote a predicted and teacher scalar price. This objective controls level error but says nothing explicit about local sensitivity error.

A structured alternative is the joint risk. In the following expression, $\mathcal{I}$ indexes the supervised first-order Greek coordinates, $\mathcal{J}$ indexes the supervised second-order coordinates, and the nonnegative coefficients $\lambda_V, \lambda_G, \lambda_H$ set the relative emphasis on prices, first derivatives, and second derivatives:

$$\mathcal{R}_{\text{joint}}(\phi) = \lambda_V \int_\Theta |\widehat{V}_\phi - V|^2 \,\mathrm{d}\mu + \lambda_G \sum_{i\in\mathcal{I}} \int_\Theta |\partial_{\theta_i}\widehat{V}_\phi - \partial_{\theta_i} V|^2 \,\mathrm{d}\mu + \lambda_H \sum_{(i,j)\in\mathcal{J}} \int_\Theta |\partial^2_{\theta_i\theta_j}\widehat{V}_\phi - \partial^2_{\theta_i\theta_j} V|^2 \,\mathrm{d}\mu. \tag{11}$$

Thus the first term measures level error, the second term measures selected Delta-type errors, and the third term measures selected Gamma- or cross-Greek-type errors.

Proposition 4.1. Joint loss as weighted Sobolev control. Here $W^{2,2}_{\mathrm{loc}}(\Theta)$ denotes the local Sobolev class with square-integrable weak derivatives up to second order on compact subsets of Θ. Assume that $V \in W^{2,2}_{\mathrm{loc}}(\Theta)$ on the selected derivative coordinates and that $\widehat{V}_\phi$ is twice weakly differentiable on the same coordinates. Then the joint risk (11) equals a weighted L^2 discrepancy between $\widehat{V}_\phi$ and V in the selected Sobolev components. In particular, if $\mathcal{R}_{\mathrm{joint}}(\phi_n) \to 0$ for a sequence (ϕ_n), then every price, first-derivative, or second-derivative term that carries a strictly positive weight in (11) converges to zero in $L^2(\mu)$.

Proof. The statement follows directly from the definition of (11). Each term is a nonnegative squared $L^2(\mu)$ norm. Convergence of the weighted sum to zero implies convergence of every active term to zero.

The proposition is elementary and is closely related to derivative-supervised or Sobolev training ideas in machine learning [7], but it clarifies the central design principle in the present derivative-surrogate setting. Greek-regularized supervision is not merely an ad hoc penalty. It changes the approximation target from a level-only norm to a norm that directly controls whichever hedge-relevant derivatives are assigned positive weight.

4.2 Normalized losses

Because price and Greek magnitudes can differ by orders of magnitude across the training domain, direct squared losses may overweight large-scale regions and underweight small but important regions. A practical implementation may therefore use normalized residuals. In the following examples, α_V and $\alpha_{G,i}$ are small positive stabilizers that prevent division by very small teacher values:

$$\ell_V^{\mathrm{rel}}(\widehat{V}, V) = \frac{|\widehat{V} - V|^2}{\alpha_V + |V|^2}, \tag{12}$$

$$\ell_{G,i}^{\mathrm{rel}}(\partial_i \widehat{V}, \partial_i V) = \frac{|\partial_i \widehat{V} - \partial_i V|^2}{\alpha_{G,i} + |\partial_i V|^2}, \tag{13}$$

Here ∂_i denotes differentiation with respect to the ith input coordinate θ_i. The first ratio normalizes price residuals, while the second normalizes the residual in the ith supervised first derivative. More refined versions can combine absolute and relative components. The choice should be driven by task priorities. Hedging studies may place more weight on local Greek fidelity, whereas calibration studies may prefer a more balanced trade-off between price fit and derivative fit.

4.3 Direct derivative heads versus automatic differentiation

Two implementation paths are available. In the first, a single scalar network $\widehat{V}_\phi$ is differentiated by automatic differentiation to obtain $\nabla \widehat{V}_\phi$ and selected Hessian coordinates. In the second, derivative quantities are predicted by dedicated heads. The first route enforces integrability automatically because the same scalar potential generates all supervised derivatives. The second route can be cheaper computationally when many derivative labels are used, but then internal consistency should be encouraged, for example by adding agreement penalties between head outputs and the automatic derivatives of the price head on a subsample.

In the present study, both routes appear in a controlled way. Model A is a price-only benchmark, with Greeks recovered afterward by finite differences, while Models B and C use direct multi-output supervision on price, Delta, and a stabilized Gamma target. The direct-head route is therefore already active in the free-data benchmark, and it is expected to become even more useful in the more expensive DSE/NDSE settings where auxiliary structure is central rather than optional.

5 Decomposition-aware learning from DSE and NDSE

5.1 Product-form supervision

The representation (1) suggests that a network need not learn only the composite map $\theta \mapsto V(\theta)$. It may instead learn component heads. In the following display, $\widehat{a}_{k,\phi}$ denotes the learned primitive head and $\widehat{w}_{k,\phi}$ denotes the learned digital or conditional-weight head for leg $k = 1, 2$:

$$\widehat{V}_{\phi}(\theta) = \widehat{a}_{1,\phi}(\theta)\, \widehat{w}_{1,\phi}(\theta) + \widehat{a}_{2,\phi}(\theta)\, \widehat{w}_{2,\phi}(\theta), \tag{14}$$

with direct supervision on selected heads when teacher values are available. The primitive heads $\widehat{a}_{k,\phi}$ can be interpreted as smooth contract- and state-dependent factors, while the weight heads $\widehat{w}_{k,\phi}$ capture digital or conditional-law blocks.

Proposition 5.1. Deterministic product error bound. Let $V = a_1 w_1 + a_2 w_2$ and $\widehat{V} = \widehat{a}_1 \widehat{w}_1 + \widehat{a}_2 \widehat{w}_2$. Then for every input θ,

$$|V - \widehat{V}| \le |a_1|\,|w_1 - \widehat{w}_1| + |a_2|\,|w_2 - \widehat{w}_2| \tag{15}$$
$$+ |\widehat{w}_1|\,|a_1 - \widehat{a}_1| + |\widehat{w}_2|\,|a_2 - \widehat{a}_2|. \tag{16}$$

All terms in this bound are evaluated at the same input θ. Consequently, when primitive and weight heads are each controlled in a suitable norm and the complementary factors are uniformly bounded, price error is controlled by the sum of componentwise errors.

Proof. Subtract and add $a_1 \widehat{w}_1 + a_2 \widehat{w}_2$ and apply the triangle inequality.

The proposition is again simple, but it explains why intermediate supervision may matter. If the network is asked to learn only the composite price map, errors in primitive and boundary channels can offset one another in-sample while leaving unstable out-of-sample sensitivities. Direct head supervision reduces that ambiguity.

5.2 Structured multi-head loss

A decomposition-aware loss can take the form below, where $\mathcal{L}_V$, $\mathcal{L}_G$, $\mathcal{L}_H$, $\mathcal{L}_W$, and $\mathcal{L}_A$ denote loss components for prices, Greeks, second derivatives, weight heads, and primitive heads, respectively, and the corresponding λ's are nonnegative tuning weights:

$$\mathcal{L}_{\text{multi}} = \lambda_V \mathcal{L}_V + \lambda_G \mathcal{L}_G + \lambda_H \mathcal{L}_H + \lambda_W \mathcal{L}_W + \lambda_A \mathcal{L}_A, \tag{17}$$

In a DSE setting, the most natural auxiliary targets are digital weights and conditional-density-type blocks. In an NDSE setting, distinct heads may be aligned to the spot-leg and strike-leg measures. This architecture is especially attractive when training labels are generated offline and stored once, because the intermediate targets then impose little additional runtime cost during training.

5.3 When the product form should be used

The product form is not mandatory. It is most attractive when three conditions are met:

1. the teacher already provides interpretable intermediate quantities;
2. the boundary channel is numerically difficult or statistically sparse;
3. out-of-sample stability matters more than minimal architectural simplicity.

In a clean BSM experiment, a single scalar network may suffice. In a stochastic-discount digital-heavy setting, decomposition-aware heads may be materially more useful.

6 Boundary-aware learning from DBSE

6.1 Why boundaries matter

For digital blocks of the form (2), first differentiation yields weighted Dirac terms and second differentiation yields δ'-type terms. These objects are concentrated near the exercise boundary. Accordingly, the parts of the input space that most strongly influence hedge-relevant derivatives may occupy a small geometric volume under naive sampling. A price-only empirical risk therefore tends to underemphasize them.

This mismatch creates a statistical problem rather than only a numerical one. A network with limited approximation capacity naturally allocates capacity where the empirical objective places mass. If the loss is dominated by smooth regions far from the boundary, then the training process may never receive enough signal to resolve the singular layer well.

6.2 Mollified labels

A direct use of Dirac targets in finite-sample learning is unattractive. DBSE instead motivates mollified approximations. Let ρ be a smooth kernel with unit mass, let z denote the scalar boundary residual such as $X - m$, and let $\varepsilon > 0$ be a bandwidth. Define

$$\rho_\varepsilon(z) = \frac{1}{\varepsilon}\rho\left(\frac{z}{\varepsilon}\right). \tag{18}$$

The scaled kernel ρ_ε concentrates around zero as ε decreases. Thus a Dirac block $\delta(X - m)$ is approximated by $\rho_\varepsilon(X - m)$, and derivative blocks can be approximated by corresponding kernel derivatives when numerically stable. During training, the bandwidth can be scheduled as

$$\varepsilon_0 > \varepsilon_1 > \cdots > \varepsilon_K > 0, \tag{19}$$

where the subscript $k = 0, \ldots, K$ denotes the training stage. The schedule starts from a smoother target and proceeds gradually toward a sharper one. This continuation strategy balances bias and stability.

Remark 6.1. The bandwidth schedule is part of the statistical design rather than an approximation afterthought. Large initial bandwidths stabilize gradients and make the auxiliary task learnable. Later bandwidth reductions restore local sharpness once the network has acquired a reasonable coarse representation.

6.3 Adaptive boundary sampling

Let $d(\theta)$ denote a distance-to-boundary proxy. In a plain call/put setting, one may take normalized log-moneyness. In an NDSE setting, a natural choice is the transformed boundary variable induced by $\widetilde{E}_{t,T} - \widetilde{m}$. Define a nonnegative boundary emphasis function $\omega(d)$, for example

$$\omega(d) = 1 + \kappa \exp\left(-\frac{d^2}{2\sigma_d^2}\right), \tag{20}$$

with parameters $\kappa, \sigma_d > 0$, where κ controls the height of the boundary emphasis and σ_d controls its width in the distance coordinate. Sampling may then be performed from a reweighted design distribution

$$\mu_{\mathrm{bdry}}(\,\mathrm{d}\theta) \propto \omega(d(\theta))\, \mu_0(\,\mathrm{d}\theta), \tag{21}$$

where μ_0 is a baseline design distribution and the proportionality constant normalizes μ_{bdry} into a probability distribution.

Proposition 6.1. Importance-corrected risk under adaptive sampling. Let μ_{bdry} be absolutely continuous with respect to μ_0 and let $r(\theta) = \frac{\mathrm{d}\mu_0}{\mathrm{d}\mu_{\mathrm{bdry}}}(\theta)$. Then, for any nonnegative measurable loss integrand f,

$$\int f(\theta)\, \mu_0(\,\mathrm{d}\theta) = \int r(\theta) f(\theta)\, \mu_{\mathrm{bdry}}(\,\mathrm{d}\theta). \tag{22}$$

Here $r(\theta)$ is the Radon–Nikodym correction that converts expectations under the boundary-enriched distribution back to expectations under the baseline distribution. Thus, boundary-enriched sampling may be combined with importance weighting to preserve the original target risk under μ_0.

Proof. The statement is the Radon–Nikodym identity.

This observation makes boundary oversampling statistically legitimate even when the target metric is defined under a different base distribution.

6.4 Auxiliary digital heads

A further refinement is to include a dedicated digital or boundary head. Here $W(\theta)$ is the teacher-supplied auxiliary boundary target and $\widehat{W}_\phi(\theta)$ is the corresponding model output:

$$\widehat{W}_\phi(\theta) \approx W(\theta), \tag{23}$$

for selected digital probabilities, conditional densities, or mollified blocks. Shared hidden layers then transfer boundary information to the price and Greek heads. This design is especially attractive when short maturity or near-the-money performance is a primary concern.

6.5 A boundary-aware hypothesis

The boundary-aware hypothesis articulated in DBSE is that price-only supervision is weakest exactly where payoff geometry becomes locally singular or nearly singular, and that a surrogate trained with boundary-sensitive labels or sampling should therefore gain most in the short-dated near-the-money region. In the present paper, that hypothesis is tested in a deliberately modest form. Boundary awareness is introduced through sample weighting rather than through a full digital-head or mollifier-head architecture. The purpose is not to claim that every additional boundary layer must dominate every metric, but to test whether a carefully tuned boundary emphasis can improve the risk-relevant local geometry of the surrogate.

7 Numeraire-aware learning under stochastic discounting

7.1 Measure-aware supervision

In NDSE, different legs of the price are naturally associated with different equivalent measures. This suggests that a learning system should not collapse all supervision into a single undifferentiated target family when rate dynamics matter. Instead, labels can be organized by leg and by measure. A decomposition-aware network may therefore include separate spot-leg and strike-leg channels, with auxiliary heads aligned to $\mathbb{Q}^+$- and $\mathbb{Q}^T$-type weights.

This design serves two purposes. First, it preserves the economic interpretation already present in the teacher. Second, it offers a way to separate smooth primitive dependence on the curve state from sharper dependence induced by the exercise boundary.

7.2 Curve encoding

A practical stochastic-discount surrogate requires a compact representation of the term structure. Several low-dimensional encodings are possible:

a) principal-component coordinates of the zero-coupon curve,

b) factor coordinates from an HJM or LMM specification,

c) a small grid of benchmark zero rates or forward rates,

d) a bond-price basis at selected maturities.

The preferred choice should reflect the label generator. When the teacher is an HJM-style engine, factor coordinates offer the most natural alignment. When training is based on a generic stochastic-rate simulation, curve PCA is often simpler.

7.3 Generalization question

A core claim to test is whether numeraire-aware supervision improves robustness under rate shifts. Let μ_{train} denote a training distribution over curve states and $\mu_{\mathsf{test}}^{\mathrm{shift}}$ a shifted distribution with altered slope, level, or curvature. The relevant question is whether a surrogate trained on structured leg- and measure-aware targets degrades less than a surrogate trained only on aggregate prices. That question is particularly important for long-dated options, callable structures approximated by European proxies, and risk systems that must operate across changing macro environments.

Remark 7.1. The advantage of a numeraire-aware design is not expected to arise in every task. The effect should be strongest when discounting is materially stochastic over the maturities of interest or when rate-curve shifts interact nontrivially with the exercise boundary.

8 A unified training framework

8.1 Structured objective

The preceding sections suggest a unified objective of the form below. The trainable parameter is ϕ, and all coefficients $\lambda_V, \lambda_G, \lambda_H, \lambda_W, \lambda_A, \lambda_C, \lambda_{\text{arb}}$ are nonnegative loss weights:

$$\mathcal{L}(\phi) = \lambda_V \mathcal{L}_V + \lambda_G \mathcal{L}_G + \lambda_H \mathcal{L}_H + \lambda_W \mathcal{L}_W + \lambda_A \mathcal{L}_A \tag{24}$$

$$+ \lambda_C \mathcal{L}_{\text{cons}} + \lambda_{\text{arb}} \mathcal{L}_{\text{arb}}, \tag{25}$$

where

- $\mathcal{L}_V$ supervises prices,
- $\mathcal{L}_G$ supervises selected first-order Greeks,
- $\mathcal{L}_H$ supervises selected second-order Greeks,
- $\mathcal{L}_W$ supervises digital or boundary-aware auxiliary heads,
- $\mathcal{L}_A$ supervises decomposition primitives when available,
- $\mathcal{L}_{\text{cons}}$ enforces consistency between direct heads and automatic derivatives or between composite and decomposed outputs,
- $\mathcal{L}_{\text{arb}}$ penalizes obvious static-arbitrage violations such as incorrect monotonicity or convexity in selected coordinates.

A representative consistency penalty is shown below. The expectation is taken over the training or design distribution, $\widehat{G}_{\phi,i}$ is the direct head for the ith Greek, and the second term checks whether the decomposed heads recombine to the price head:

$$\mathcal{L}_{\text{cons}} = \mathbb{E}\Big[\big|\widehat{G}_{\phi,i} - \partial_{\theta_i} \widehat{V}_\phi\big|^2\Big] + \mathbb{E}\Big[\big|\widehat{V}_\phi - (\widehat{a}_{1,\phi}\widehat{w}_{1,\phi} + \widehat{a}_{2,\phi}\widehat{w}_{2,\phi})\big|^2\Big], \tag{26}$$

In the first expectation, i is a supervised Greek coordinate. In the second expectation, the decomposed primitive and weight heads are evaluated at the same input as the price head. The formula is modified in the obvious way when only a subset of heads is active.

8.2 Training protocol

A training protocol consistent with the engine hierarchy can be organized as follows.

1. Generate a base design sample from a broad domain of contracts, model parameters, and curve states.
2. Mark near-boundary samples using a proxy $d(\theta)$ and enrich them by oversampling or by increased loss weight.
3. Compute price labels everywhere and Greek or boundary labels where the engine is reliable enough to do so.
4. Apply numerical safeguards for scale, step-size, and stability before derivative labels or finite-difference diagnostics are used.
5. Begin training with smoothed auxiliary boundary targets and moderate derivative weights.
6. Gradually sharpen the boundary bandwidth and raise derivative or auxiliary weights as the fit stabilizes.
7. Evaluate both globally and conditionally on challenging regions such as short-dated, near-the-money, or rate-shifted slices.

This staged procedure aligns naturally with the structure of the teacher engines and avoids presenting all difficult targets at full sharpness from the first epoch.

8.3 Pseudo-algorithm

Algorithm 1. Sensitivity-engine-guided surrogate training

Input: design distribution μ_0, boundary proxy $d(\theta)$, teacher engine $\mathcal{E}$, bandwidth schedule (ε_k), architecture class $\{\widehat{V}_\phi, \widehat{G}_\phi, \widehat{H}_\phi, \widehat{W}_\phi\}$, loss weights λ.
1. Sample base inputs $\theta_n \sim \mu_0$ and form a boundary-enriched sample when $|d(\theta_n)|$ is small.
2. Use the teacher engine to compute prices and available auxiliary labels (V, G, H, W, A).
3. Initialize the network under a smooth auxiliary regime with bandwidth ε_0.
4. For each training stage k, minimize the objective (25) with boundary bandwidth ε_k and current weights $\lambda^{(k)}$.
5. Monitor global metrics and conditional metrics on boundary and rate-shift slices.
6. Select the final model using a validation score that combines level, Greek, and local robustness criteria.
Output: trained surrogate and associated diagnostic metrics.

9 Interpretation of structured supervision

The framework suggests four main advantages, each with a corresponding qualification.

First, additional supervisory channels can improve sample efficiency when the goal is not merely price interpolation but derivative fidelity. If the teacher can provide stable Greek labels, part of the local geometry that would otherwise need to be inferred indirectly becomes directly observable during training.

Second, direct supervision on derivatives and boundary-related targets can improve local robustness in regions where a price-only loss is weak. This is the main mechanism examined in the present paper. The empirical study later shows that this local-geometry channel is economically meaningful even when the gains are not driven by a dramatic headline change in average price RMSE alone.

Third, decomposition-aware and measure-aware supervision can improve interpretability and may improve robustness when discounting is stochastic or when the teacher itself has a structured two-leg representation. That broader advantage is part of the framework set out here even though it is not fully tested in the current free-data implementation.

Fourth, structured intermediate heads can make model diagnosis more transparent. When a surrogate underperforms, the error can sometimes be traced to a particular component, such as price level, first derivative, second derivative, or boundary-sensitive auxiliary block, rather than being treated as an undifferentiated black-box failure.

These advantages do not imply that more structure is always better. Extra labels increase offline generation cost. Higher-order labels may be noisy in Monte Carlo settings if they are not carefully estimated. Multi-head decompositions can introduce tuning choices that weaken performance when auxiliary tasks are poorly balanced. The practical question is therefore not whether structured supervision dominates in every dimension, but which structure best matches the intended use of the surrogate.

10 Empirical study design and results

10.1 Position of the present study within the broader program

The broader empirical program implied by the present framework has several layers. These include exact-label baseline experiments, deterministic-discount non-BSM experiments, and stochastic-discount numeraire-aware experiments. The present paper implements the first of these layers in completed form. The study uses public daily market-state data, generates synthetic teacher labels from a BSM engine, and evaluates whether structured supervision improves price fit, Greek accuracy, and hedging performance relative to a price-only benchmark. This design is intentionally narrower than the full DSE/NDSE agenda, but it is sufficient to test the paper's most immediate empirical claim: sensitivity-guided supervision should matter most in local risk-sensitive regions rather than only through headline average-fit improvements. It also remains close to the derivative-surrogate and deep-hedging literature in which fast approximation and hedge-aware objectives are central practical concerns [8, 9].

10.2 Data construction and teacher labels

The implementation uses daily public data loaded in local-file mode. The underlying index level is the daily S&P 500 close from Stooq [11]. The term-structure proxies are the FRED constant-maturity Treasury series DGS3MO, DGS2, DGS10 [12], and the implied-volatility proxies are VIXCLS and VXVCLS, interpreted as approximately 30-day and 90-day implied-volatility levels [13, 14]. The script first computes S&P 500 log returns and then constructs realized-volatility features. In the following display, S_t is the S&P 500 close at trading date t, $\Delta \log S$ is the daily log return, and $\mathrm{sd}(\cdot)$ is the rolling sample standard deviation over the indicated window:

$$RV_{20,t} = \sqrt{252}\,\mathrm{sd}(\Delta \log S_{t-19:t}), \qquad RV_{60,t} = \sqrt{252}\,\mathrm{sd}(\Delta \log S_{t-59:t}),$$

The factor $\sqrt{252}$ annualizes the rolling daily standard deviation. These realized-volatility proxies are computed on the pure S&P 500 trading-day series before merging the FRED data. After the merge, the macro and volatility series are forward-filled on S&P 500 trading days, and the implied-volatility inputs are set as decimal volatility proxies:

$$IV_{30,t} = \mathrm{VIXCLS}_t/100, \qquad IV_{90,t} = \mathrm{VXVCLS}_t/100.$$

The cleaned daily state contains 4,601 trading dates from 2007-12-04 through 2026-03-19.

Teacher labels are generated by a standalone BSM call engine with dividend yield fixed at $q = 0$, using the classical pricing framework of BSM as the exact teacher [1, 2]. For each date t, the maturity-specific rate $r_t(\tau)$ is obtained by piecewise-linear interpolation and end-point extrapolation across the maturity anchors 0.25, 2, and 10 years using $\mathrm{DGS3MO}_t$, $\mathrm{DGS2}_t$, and $\mathrm{DGS10}_t$. The implied-volatility proxy $IV_t(\tau)$ is obtained by piecewise-linear interpolation and end-point extrapolation between 30/365 and 90/365 years. The teacher volatility is then defined by the following maturity-dependent blend, where $\sigma_t(\tau)$ is the BSM teacher volatility at date t and maturity τ, $IV_t(\tau)$ is the interpolated implied-volatility proxy, and $\alpha(\tau)$ is the interpolation weight:

$$\sigma_t(\tau) = \alpha(\tau) IV_t(\tau) + \big(1 - \alpha(\tau)\big) RV_{20,t}, \qquad \alpha(\tau) = \begin{cases} 0.8, & \tau \le 30/365, \\ 0.6, & \tau > 30/365. \end{cases}$$

The synthetic option panel is built on the fixed grids below. Here τ is time to maturity in years and k is log-moneyness:

$$\begin{aligned} &\tau \in \{3, 5, 7, 14, 21, 30, 45, 60, 90\}/365, \\ &k \in \{-0.15, -0.10, -0.07, -0.05, -0.03, -0.02, -0.01, 0, 0.01, 0.02, 0.03, 0.05, 0.07, 0.10, 0.15\}. \end{aligned}$$

with strike $K = S_t e^k$, so positive k corresponds to an out-of-the-money call strike above spot and negative k corresponds to an in-the-money call strike below spot. The local boundary region used for the paper's local metrics and the hedging study is

$$\mathcal{B} = \{(t, \tau, k) : \ \tau \le 30/365, \ |k| \le 0.05\}.$$

Thus $\mathcal{B}$ contains short-dated contracts with maturity at most 30 calendar days and log-moneyness within five percentage points of the money. The model input vector is

$$x_t(\tau, k) = \big(k, \tau, r_t(\tau), RV_{20,t}, RV_{60,t}, IV_{30,t}, IV_{90,t}, IV_{90,t} - IV_{30,t}\big).$$

The final coordinate, $IV_{90,t} - IV_{30,t}$, is a simple implied-volatility slope proxy.

The split is calendar based rather than randomly shuffled. Training dates run through 2017-12-31, validation dates through 2021-12-31, and the test set contains all later dates. Because the grid is fixed across dates, the split counts are large and transparent. Table 1 reports the resulting panel sizes.

Table 1: Sample counts from the completed empirical run

Split	Total samples	Boundary samples	Dates
Train	342,495	136,998	2,537
Validation	136,080	54,432	1,008
Test	142,560	57,024	1,056

10.3 Surrogate specifications, training, and model selection

All three surrogates are implemented as two-hidden-layer multilayer perceptrons with width 64 per layer, ReLU activation, Adam optimization [10], learning rate 10^{-3}, weight decay 10^{-5}, batch size 512, early stopping with validation fraction 0.1, patience 10, maximum iteration count 120, and random seed 1234. Model A is price only and recovers Δ and Γ afterward by fourth-order centered finite differences in log-moneyness. The effective finite-difference step is bounded below by a double-precision roundoff scale, so that the second derivative is not reconstructed from an excessively small price difference. The resulting derivatives with respect to k are then converted to spot derivatives using $K = Se^k$.

Models B and C use multi-output supervision on the following three heads, where C is the call price, Δ is the spot Delta, Γ is the spot Gamma, and $\lambda_\Delta, \lambda_\Gamma$ are the Delta and Gamma loss weights:

$$\left(C, \sqrt{\lambda_\Delta}\,\Delta, \sqrt{\lambda_\Gamma}\,\operatorname{asinh}(S^2\Gamma)\right),$$

Here $\operatorname{asinh}(S^2\Gamma)$ is an invertible scale stabilization of the Gamma target; multiplying the Delta and Gamma channels by $\sqrt{\lambda_\Delta}$ and $\sqrt{\lambda_\Gamma}$ implements the intended weighted squared loss. This stabilizes the Gamma channel while preserving the weighted-loss interpretation. The common Greek-loss defaults are $\lambda_\Delta = 1.0$ and $\lambda_\Gamma = 5.0$. The boundary-specific defaults used only for Model C are $\lambda_{\text{boundary}} = 4.0$ and $h = 0.02$. Model B is selected on the validation set over the grid

$$\lambda_\Delta \in \{1.0, 0.85\}, \qquad \lambda_\Gamma \in \{5.0, 3.5, 2.5\},$$

using a composite score that weights local price, global price, local Δ, and local Γ RMSE in the ratio

$$(3.0,\ 0.75,\ 1.25,\ 1.0)$$

relative to the Model A baseline, in the order just listed. The selected Model B candidate is

$$(\lambda_\Delta, \lambda_\Gamma) = (0.85, 3.5).$$

Model C further applies the boundary-aware sample weight below. In this expression, ω_i is the sample weight for observation i, k_i and τ_i are its log-moneyness and maturity, $\lambda_{\text{boundary}}$ controls the intensity of the boundary emphasis, and h controls the width of the near-money bump:

$$\omega_i = 1 + \lambda_{\text{boundary}} \exp\left(-\frac{k_i^2}{h^2}\right) \mathbf{1}_{\{\tau_i \le 30/365\}},$$

and is chosen from a broader Model C-specific search. Its validation search uses

$$\lambda_\Delta \in \{0.85, 1.0\}, \qquad \lambda_\Gamma \in \{2.0, 2.75, 3.5, 4.25\},$$

$$\lambda_{\text{boundary}} \in \{0.8, 1.4, 2.0, 2.8\}, \qquad h \in \{0.02, 0.025, 0.03\},$$

with a more price-aware selection score weighting local price, global price, local Δ, and local Γ, again in that order, as

$$(4.5,\ 1.0,\ 1.0,\ 0.75).$$

The selected Model C candidate is

$$(\lambda_\Delta, \lambda_\Gamma, \lambda_{\text{boundary}}, h) = (1.0, 4.25, 0.8, 0.02).$$

Thus Model B remains on the balanced selection path described above, while Model C uses a softer boundary weighting than a more aggressive boundary-weighted specification.

10.4 Numerical accuracy controls

The empirical design is also a numerical-computing problem. The teacher labels, stabilized targets, finite-difference Greeks, and validation metrics are affected by roundoff, truncation error, scaling, and conditioning. The implementation therefore uses double precision as the default arithmetic layer, a vectorized stable normal CDF in the BSM teacher, and a log-ratio construction for the BSM input. In the following expression, S is spot, K is strike, r is the continuously compounded rate, σ is volatility, and τ is time to maturity:

$$d_1 = \frac{\log S - \log K + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$

Here d_1 is the usual BSM standardized log-moneyness input to the normal CDF. The implementation also uses scale-normalized training targets for the structured Gamma head. These choices do not change the mathematical teacher, but they reduce avoidable numerical noise when a large panel of short-maturity near-the-money options is generated.

The price-only benchmark requires special care because its Greeks are reconstructed from a learned price surface. If $c(k)$ denotes the predicted price as a function of log-moneyness while all other state variables are held fixed, and if h_{fd} denotes the finite-difference step used only for numerical reconstruction, the code uses the fourth-order centered stencils

$$c_k(k) \approx \frac{-c(k+2h_{\text{fd}}) + 8c(k+h_{\text{fd}}) - 8c(k-h_{\text{fd}}) + c(k-2h_{\text{fd}})}{12h_{\text{fd}}},$$

and

$$c_{kk}(k) \approx \frac{-c(k+2h_{\text{fd}}) + 16c(k+h_{\text{fd}}) - 30c(k) + 16c(k-h_{\text{fd}}) - c(k-2h_{\text{fd}})}{12h_{\text{fd}}^2}.$$

The derivatives c_k and c_{kk} are derivatives with respect to log-moneyness, not directly with respect to spot. Because the grid uses $K = Se^k$, the spot Greeks are then obtained from the identities

$$\Delta = -\frac{c_k}{S}, \qquad \Gamma = \frac{c_{kk} + c_k}{S^2}.$$

In these identities, Δ and Γ denote derivatives with respect to spot S, whereas c_k and c_{kk} are derivatives with respect to log-moneyness k. For a fourth-order second derivative, the truncation error is of order h_{fd}^4, while roundoff is amplified on the order of $\epsilon_{\text{mach}}/h_{\text{fd}}^2$, where ϵ_{mach} denotes machine epsilon for the floating-point arithmetic. The implementation therefore uses an effective step no smaller than an $\epsilon_{\text{mach}}^{1/6}$-scale floor. This safeguard follows the standard numerical-analysis tradeoff between truncation error and propagated roundoff error [6].

These controls are intentionally modest. They are not presented as a separate numerical-analysis contribution. Their purpose is to keep the empirical comparison focused on the statistical supervision question rather than on avoidable artifacts from unstable differencing, poorly scaled targets, or tail-probability evaluation.

10.5 Evaluation metrics

The empirical evaluation is intentionally derivative oriented. The primary global metrics are test-set RMSE for price, Delta, and Gamma over the full synthetic option panel. The primary local metrics are the same RMSE quantities restricted to the boundary region $\mathcal{B}$. Economic usefulness is then assessed through a five-day discrete hedging experiment in the same local region. The hedging table reports mean P&L, standard deviation of P&L, the 95% quantile of absolute P&L, and an expected-shortfall-style loss measure. The central empirical question is not whether one model dominates every metric, but whether structured supervision improves local risk representation in the region where hedge quality is most sensitive.

10.6 Global and local accuracy results

The completed empirical study yields a fairly clear ranking among the three specifications. Table 2 reports the global and local test-set metrics. Model A, the price-only benchmark, is uniformly the weakest specification. Model B remains a strong Greek-regularized surrogate: relative to Model A it lowers global price RMSE from 192.294 to 189.542, local price RMSE from 84.376 to 80.601, local Delta RMSE from 0.4148 to 0.0549, and local Gamma RMSE from 0.001933 to 0.001123. These are economically meaningful gains and they indicate that joint price-Greek supervision improves the local geometry of the surrogate.

A central empirical finding is that Model C is no longer merely an alternative with stronger first-order behavior. Within the present experiment, it attains the most favorable results on global price RMSE, global Delta RMSE, local price RMSE, and local Delta RMSE. In particular, Model C attains global price RMSE 186.478, local price RMSE 79.083, global Delta RMSE 0.0215, and local Delta RMSE 0.0221. Relative to Model A, the boundary-aware specification improves local first-order accuracy and average price fit. Relative to Model B, Model C improves both global and local price fit and sharpens Delta accuracy. Model B, however, still retains the best Gamma fit: its global and local Gamma RMSE are 7.49×10^{-4} and 1.123×10^{-3}, compared with 9.76×10^{-4} and 1.422×10^{-3} for Model C.

Table 2: Global and local test-set metrics from the completed empirical run

Model	Gbl. price	Gbl. Δ	Gbl. Γ	Loc. price	Loc. Δ	Loc. Γ
Model A	192.294226	0.441581	0.001389	84.375994	0.414776	0.001933
Model B	189.541986	0.043754	0.000749	80.601021	0.054865	0.001123
Model C	186.477788	0.021536	0.000976	79.082853	0.022096	0.001422

These results should be interpreted within the limited scope of the present empirical design. The central claim is not that every structured surrogate must dominate every alternative on every statistic, but that boundary-aware sensitivity supervision should improve behavior where local risk is concentrated. On that criterion, the present run is supportive, though still limited to the present design. Model C improves local price and local Delta materially, and it also improves local Gamma relative to the price-only benchmark. The remaining Model B advantage on Gamma is best read as a model-selection tradeoff rather than a contradiction of the boundary-aware result.

10.7 Hedging results and empirical interpretation

The complementary economic evaluation is based on the five-day discrete hedging experiment in the local boundary region and is reported in Table 3. The hedging experiment is populated with 28,377 valid local contracts for each model. All three models have negative mean P&L relative to the teacher benchmark, so the economically more informative comparison is the dispersion and tail behavior rather than the sign of the mean alone. Model A has the widest hedge-error distribution, with standard deviation 74.52, 95% absolute P&L quantile 219.73, and expected-shortfall-style loss 277.76. Model B materially improves all three statistics, reducing the standard deviation to 63.35 and the expected-shortfall-style loss to 235.95. Model C improves them further, with standard deviation 62.66, 95% absolute P&L quantile 188.28, and expected-shortfall-style loss 229.46. Thus the boundary-aware specification improves local statistical fit and, within the present experiment, yields the most stable hedge distribution among the three models.

Table 3: Five-day local-boundary hedging metrics from the completed empirical run

Model	Mean P&L	Std. dev.	95% abs. quantile	ES_{95} loss	Contracts
Model A	-68.571048	74.521978	219.732340	277.755591	28,377
Model B	-64.057901	63.348486	191.938608	235.954129	28,377
Model C	-61.621267	62.655553	188.277232	229.464071	28,377

The empirical conclusion can now be stated more directly. Model A is the weakest price-only baseline. Model B provides a strong Greek-regularized reference point. Model C is the most favorable overall empirical specification within the completed study: it preserves the price and Delta advantages of structured supervision, improves local price fit and hedging stability in the boundary region, and does so without sacrificing global accuracy. The evidence therefore suggests that boundary-aware Greek-regularized supervision improves derivative-surrogate behavior primarily through local risk fidelity and hedge robustness.

10.8 Relation to the broader empirical agenda

The present study is only the first completed empirical layer of the broader framework. It validates the usefulness of Greek-regularized supervision in an exact-label environment and gives a stronger test of boundary weighting than the earlier draft. The next empirical step is to move from a BSM teacher to a deterministic-discount non-BSM teacher aligned with DSE, where decomposition-aware heads and auxiliary digital targets can be tested directly rather than indirectly through a BSM-based proxy. A further step is the stochastic-discount setting aligned with NDSE, where the main question is no longer only local price and Greek fidelity, but robustness under rate or curve shifts when the supervision itself is measure aware.

11 Discussion and limitations

The framework set out here is intentionally derivative focused. It is not intended as a general solution for return prediction, asset allocation, market microstructure forecasting, or generic financial tabular learning. A natural use case is a setting in which the target function is already defined by a pricing model or a high-quality numerical engine and the practical problem is to compress that target into a low-latency surrogate without losing local risk structure.

The empirical evidence should therefore be interpreted with discipline. The present study uses a synthetic BSM teacher rather than direct market option quotes, so it illustrates the value of structured supervision under controlled teacher labels rather than full market microstructure realism. The data inputs are free public market-state proxies, not a complete option panel. The boundary region is hand-defined by short maturity and near-the-money moneyness, which is appropriate for the paper's objective but still design dependent. The hedging experiment is discrete and local, so it is informative about comparative robustness but not a complete trading study.

Several broader limitations remain. First, the framework inherits the assumptions and modeling choices of the teacher engine. If the teacher is misspecified, supervised learning may reproduce that misspecification efficiently rather than correct it. Second, additional labels are useful only when their own numerical error is controlled. This is especially important for second-order quantities in Monte Carlo settings. Third, the current empirical implementation covers European-style calls in a BSM environment; path dependence, early exercise, and high-dimensional callable structures would require additional work. Fourth, the present results still caution against treating every additional layer of structure as automatically beneficial. The practical lesson is comparative rather than absolute. Structured supervision should be chosen to match the intended risk task.

12 Conclusion

This paper presents a derivative-surrogate learning framework built on an earlier sensitivity-engine sequence and studies that framework in a first completed empirical implementation. BSE supplies exact baseline labels in the BSM setting. DSE supplies model-agnostic digital and conditional-law blocks under deterministic discounting. NDSE extends those structures to stochastic discounting through numeraire-matched measures. DBSE isolates the boundary layer through Dirac, δ', mollifier, and likelihood-ratio components. Taken together, these ingredients support learning systems trained not only on prices but also on sensitivities, intermediate probabilistic blocks, and boundary-aware auxiliary targets.

The paper's methodological claim is intentionally limited. The framework does not replace pricing theory or classical numerical methods. Its role is to transfer pricing structure into statistical objectives and architectures, especially in applications where stable first- and second-order sensitivities matter as much as level fit. The present free-data empirical study supports that claim in a concrete way. Relative to a price-only baseline, structured supervision improves Greek accuracy and reduces local hedging dispersion and tail loss. Within the completed study, the boundary-aware Greek-regularized surrogate, Model C, gives the most favorable overall empirical performance, while the Model B Gamma advantage confirms that the ranking remains metric dependent.

The resulting conclusion is therefore specific rather than broad. The value of sensitivity-engine-guided learning in the present implementation appears to lie primarily in local risk fidelity and hedge robustness rather than in broad average-fit claims. The empirical evidence also clarifies the role of boundary-aware design: it appears most reliable when tuned carefully rather than when boundary emphasis is imposed too aggressively. Subsequent empirical layers can test the decomposition-aware and measure-aware components of the framework more directly under non-BSM and stochastic-discount teachers.

References

[1] Black, F., and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.

[2] Merton, R. C. (1973). Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1), 141–183.

[3] Geman, H., El Karoui, N., and Rochet, J.-C. (1995). Changes of numeraire, changes of probability measure and option pricing. *Journal of Applied Probability*, 32(2), 443–458.

[4] Broadie, M., and Glasserman, P. (1996). Estimating security price derivatives using simulation. *Management Science*, 42(2), 269–285.

[5] Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.

[6] Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms*. Second edition. SIAM.

[7] Czarnecki, W. M., Osindero, S., Jaderberg, M., Swirszcz, G., and Pascanu, R. (2017). Sobolev training for neural networks. *Advances in Neural Information Processing Systems*, 30.

[8] Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019). Deep hedging. *Quantitative Finance*, 19(8), 1271–1291.

[9] Horvath, B., Muguruza, A., and Tomas, M. (2021). Deep learning volatility: a deep neural network perspective on pricing and calibration in (rough) volatility models. *Quantitative Finance*, 21(1), 11–27.

[10] Kingma, D. P., and Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations*.

[11] Stooq. (2026). S&P 500 Index historical daily data. Retrieved March 20, 2026, from `https://stooq.com/q/d/?s=%5Espx`.

[12] Board of Governors of the Federal Reserve System (US). (2026). Market yield on U.S. Treasury securities at constant maturity, quoted on an investment basis: DGS3MO, DGS2, and DGS10. Retrieved from FRED, Federal Reserve Bank of St. Louis. Retrieved March 20, 2026, from `https://fred.stlouisfed.org/`.

[13] Chicago Board Options Exchange. (2026). CBOE Volatility Index: VIX [VIXCLS]. Retrieved from FRED, Federal Reserve Bank of St. Louis. Retrieved March 20, 2026, from `https://fred.stlouisfed.org/series/VIXCLS`.

[14] Chicago Board Options Exchange. (2026). CBOE S&P 500 3-Month Volatility Index [VXVCLS]. Retrieved from FRED, Federal Reserve Bank of St. Louis. Retrieved March 20, 2026, from `https://fred.stlouisfed.org/series/VXVCLS`.

[15] Shen, D. H. (2026). An Implementation-Friendly Derivation of First- and Second-Order Black–Scholes–Merton Sensitivities (January, 2026). SSRN Working Paper. Available at SSRN: https://ssrn.com/abstract=6089246. DOI: http://dx.doi.org/10.2139/ssrn.6089246.

[16] Shen, D. H. (2026). A Distributional Sensitivity Engine: Greeks from Two-Measure Digital Weights and Conditional Law Blocks (February 2026). SSRN Working Paper. Available at SSRN: https://ssrn.com/abstract=6172183. DOI: http://dx.doi.org/10.2139/ssrn.6172183.

[17] Shen, D. H. (2026). Numeraire-Based Distributional Sensitivity Engine for European Options under Stochastic Interest Rates (March, 2026). SSRN Working Paper. Available at SSRN: https://ssrn.com/abstract=6438419. DOI: http://dx.doi.org/10.2139/ssrn.6438419.

[18] Shen, D. H. (2026). Dirac Blocks for Sensitivity Engines: Boundary Calculus and Likelihood-Ratio Reweighting for Boundary-Sensitive Monte Carlo Greeks (April, 2026). SSRN Working Paper. Available at SSRN: https://ssrn.com/abstract=6622718. DOI: http://dx.doi.org/10.2139/ssrn.6622718.